
Toward Early Alzheimer’s Risk Detection in PMOS (PCOS) Patients: A Multi-Tissue, Multi-Method Gene Expression Investigation

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Abstract

Polyendocrine Metabolic Ovarian Syndrome (PMOS, formerly Polycystic Ovary Syndrome, PCOS) and Alzheimer’s disease (AD) share proposed mechanistic links through insulin resistance, chronic inflammation, and oxidative stress. This report investigates whether PCOS gene expression carries an individual-level signal for AD risk, with the goal of informing early clinical monitoring. An initial machine learning classification approach produced a degenerate result, which was systematically diagnosed as a cross-platform normalization artifact rather than a real finding. After correcting this issue and independently replicating the analysis in a tissue- and platform-matched design (blood, rather than brain vs. adipose), the classifier itself was shown to be valid, yet still detected no individual-level PCOS signal across two tissues and two independent projection methods. In contrast, rank-based pathway enrichment analysis revealed a specific, replicated, population-level signal: immune and inflammatory pathways are consistently suppressed across two independent PCOS cohorts (adipose and blood), in contrast to their elevation in AD brain tissue. This report documents the complete investigation, including the failure diagnosis, the validation methodology, the final results, and their limitations, and argues that population-level mechanistic hypotheses – rather than individual-level risk classifiers – are the more tractable near-term contribution given current public data constraints.

1 Introduction

Polycystic Ovary Syndrome (PCOS) is a common endocrine disorder affecting an estimated 8–13% of women of reproductive age, characterized by hormonal imbalance, insulin resistance, and chronic low-grade inflammation [1]. In May 2026, an international consensus process renamed the condition Polyendocrine Metabolic Ovarian Syndrome (PMOS), reflecting a broader recognition of its systemic, rather than purely reproductive, character [13]. Independently, PCOS has been epidemiologically and mechanistically linked to elevated long-term risk of metabolic and cardiovascular disease, and a growing body of literature has proposed that the same underlying processes – insulin resistance,

oxidative stress, and chronic inflammation – may also elevate long-term risk of neurodegenerative disease, including Alzheimer’s disease (AD) [2].

If true, this connection would have real clinical value: a detectable molecular signal in PCOS patients could motivate earlier cognitive monitoring and precautionary intervention, potentially years before clinical AD symptoms emerge. This report investigates that specific, clinically motivated question – *can PCOS gene expression signal individual-level AD risk using public transcriptomic data?* – through a sequence of increasingly rigorous computational experiments.

The investigation proceeded in two broad phases. In the first phase, we implemented a direct machine-learning approach: train a classifier to distinguish AD from healthy controls using brain gene expression, then project PCOS patient samples through that classifier to obtain an individual-level “AD-likeness” score. This approach initially produced a clearly degenerate result. Rather than treating this as a dead end, we diagnosed the failure systematically, identified its root cause, corrected it, and then deliberately stress-tested the corrected approach using an independent, tissue-matched replication (blood tissue on both sides of the comparison, rather than brain vs. adipose). This is, to our knowledge, a more rigorous validation design than has been reported in prior work applying similar classifier-projection approaches.

In the second phase, having established that individual-level classification does not detect a reliable signal regardless of tissue, platform, or projection method, we pivoted to a population-level question better suited to small, publicly available cohorts: does rank-based pathway enrichment reveal a shared or divergent biological mechanism between PCOS and AD? This analysis, replicated across two independent PCOS datasets in different tissues, produced a specific, non-trivial, and biologically interpretable finding.

This report documents the full investigation – including the initial failure, its diagnosis, the corrected methodology, the tissue-matched replication, and the final pathway-level results – with the goal of representing both what was learned about the specific research question and about the broader methodological pitfalls of applying classifier-projection approaches to small, cross-platform public genomics data.

2 Related Work

The mechanistic overlap between PCOS and AD has been argued primarily through shared risk factor pathways rather than direct molecular comparison. Condorelli et al. [1] describe the progression from PCOS-associated insulin resistance to pancreatic β -cell dysfunction, paralleling mechanisms implicated in AD as a proposed “type 3 diabetes.” Reddy et al. [2] review the role of oxidative stress as a converging mechanism in both diabetes and AD pathogenesis. Neither study performs direct computational comparison of PCOS and AD transcriptomic data.

On the computational side, Li et al. [3] performed a differential expression and pathway enrichment analysis of PCOS using pooled public microarray datasets from the Gene Expression Omnibus (GEO), explicitly noting that individual PCOS datasets are too small (typically $n < 20$) to support robust standalone analysis – a limitation directly relevant to, and consistent with, our own findings. Lee and Lee [4] conducted the most directly comparable prior work: a systematic evaluation of blood-based gene expression classifiers for AD across three public cohorts (ADNI, AddNeuroMed ANM1/GSE63060, and ANM2/GSE63061), reporting internal cross-validation AUCs of 0.657–0.874 depending on cohort and feature selection method, and explicitly using ComBat for cross-dataset batch correction. Our blood-based classifier (Section 5) achieves a comparable AUC (0.755) using the same ANM2/GSE63061 cohort, providing external validation that our classifier construction is methodologically sound.

To our knowledge, no prior published work has specifically attempted to project PCOS gene expression through an AD-trained classifier to estimate individual-level AD risk, nor has any prior work tested such a projection for robustness across multiple tissues, platforms, and projection methods,

as we do here.

3 Data

Four publicly available GEO datasets were used, spanning two diseases, two tissue types, and two microarray/RNA-seq platforms, summarized in Table 1.

Table 1: Summary of datasets used in this investigation.

Accession	Disease	Tissue	Platform	n
GSE33000	AD vs. Control	Brain (PFC)	Microarray	467 (310 AD, 157 Control)
GSE84958	PCOS vs. Control	Adipose	RNA-seq	13 (7 PCOS, 6 Control)*
GSE54248	PCOS vs. Control	Peripheral blood	Illumina HT-12 v4	8 (4 PCOS, 4 Control)
GSE63061	AD vs. Control	Peripheral blood	Illumina HT-12 v4	273 (139 AD, 134 Control)

*Original cohort was 15 subjects (8 Control, 7 PCOS); two Control samples (MO31, MO37) were excluded as technical outliers after PCA-based diagnosis (Section 4.6).

GSE33000 and GSE63061 (the AddNeuroMed ANM2 cohort) together provide well-powered AD data in two different tissues, on two different platforms, enabling a tissue-matched replication design against the two available PCOS cohorts. GSE84958 and GSE54248 are, respectively, the largest publicly available adipose and blood PCOS transcriptomic datasets identified during this investigation; both remain small by conventional standards, a limitation discussed further in Section 7.

4 Methods

4.1 Initial classification approach

Following the general approach used in prior individual-level AD classification work, we trained a logistic regression classifier on principal components (PCA, top 20 PCs) derived from AD vs. Control gene expression in GSE33000. PCOS samples from GSE84958 were then projected into the same PC space and scored by the trained classifier to obtain an individual “AD-likeness” probability, following the logic that if PCOS gene expression resembles AD gene expression, PCOS samples should score consistently higher than Control samples.

This initial attempt produced a clearly degenerate result: all PCOS samples scored near 1.0 regardless of true PCOS/Control status, indicating the classifier was not discriminating meaningfully but rather saturating. We diagnosed the cause as a *cross-platform normalization mismatch*: GSE33000 is microarray data (log2 intensity ratios) while GSE84958 is RNA-seq count data (converted to log2 counts-per-million), and the two were never brought onto a common numerical scale before projection.

4.2 Cross-platform harmonization

To correct this, gene expression matrices from both datasets were restricted to their common gene set (9,180 genes after symbol matching), and further restricted to genes with complete data in both platforms (1,261 genes survived this filter, after removing genes with missing values present in the GSE33000 processed matrix). The combined matrix was harmonized using ComBat [5], an empirical Bayes batch-effect correction method, treating dataset (platform) as the batch variable. Because AD/Control status and PCOS/Control status are not shared covariates across the two datasets, ComBat was run without a biological covariate to preserve (intercept-only design), a known limitation discussed in Section 7.

4.3 Honest performance validation

The initial classification approach in Section 4.1 evaluated classifier performance on the same data used for training, which overstates real-world generalization. To obtain an honest performance

estimate, we implemented stratified 5-fold cross-validation, with PCA refit independently within each training fold to avoid information leakage from the held-out test fold into the feature space. Out-of-fold predictions were used to compute AUC and accuracy, providing an unbiased estimate of how well the AD/Control classifier itself performs, independent of the PCOS projection question.

4.4 Tissue-matched replication

A remaining concern was that GSE33000 (brain) and GSE84958 (adipose) represent different tissues entirely, so any true or false finding could be confounded by tissue-specific expression differences rather than disease-specific ones. To address this, we repeated the entire pipeline – classifier training, ComBat harmonization, cross-validation, and projection – using GSE63061 (AD, blood) and GSE54248 (PCOS, blood), which additionally share the same microarray platform (Illumina HumanHT-12 v4, GPL10558), removing platform mismatch as a confound as well. Raw GSE54248 probe intensities were background-corrected and quantile-normalized using `neqc` [7], with a detection-confidence filter (probes retained if the manufacturer-reported detection confidence exceeded 0.95 in at least half of samples); this filter’s polarity was verified empirically, since an initial implementation incorrectly assumed the detection column encoded a p-value rather than a confidence score, which would have discarded the vast majority of real signal. GSE63061 probe-level data were mapped to gene symbols using the manufacturer’s GPL10558 annotation manifest, with duplicate probes per gene collapsed to the highest-expressed probe.

4.5 Independent projection method

As a second, model-free check on the tissue-matched projection, we computed AD and Control “centroids” (mean expression profiles) from the harmonized blood AD dataset, and classified each PCOS-blood sample by Pearson correlation distance to each centroid. This method requires no model training or cross-validation and is not subject to the same overfitting risks as a fitted classifier, providing an independent check on whether the classifier-based projection result was method-specific.

4.6 PCOS differential expression and outlier diagnosis

Prior to the classifier work, we separately conducted a standard differential expression (DE) analysis of GSE84958 using DESeq2 [8], with log fold-change shrinkage via `apeglm` to correct for instability at low sample size. An initial run identified two Control samples (MO31, MO37) as extreme outliers via principal component analysis, disproportionately driving overall variance (PC1: 40.6% of variance, dominated almost entirely by these two samples) independent of PCOS/Control status. These samples were excluded from all subsequent analyses (Table 1). Even after exclusion, DESeq2’s own diagnostics (a high proportion of Cook’s-distance outlier-flagged genes, and optimizer convergence failures during shrinkage for several top genes) indicated that single-gene DE testing was fundamentally unreliable at this sample size ($n = 13$), motivating the shift to pathway-level, rank-based analysis for all downstream PCOS work.

4.7 Pathway-level enrichment analysis

For each dataset (GSE33000, GSE84958, GSE54248), genes were ranked by their differential expression test statistic (limma moderated t -statistic for GSE33000 and GSE54248; DESeq2 Wald statistic for GSE84958) and used as input to rank-based gene set enrichment analysis (GSEA) [9] via the `fgsea` implementation [10], against the MSigDB Hallmark and KEGG (Legacy) gene set collections [11]. This approach uses the entire ranked gene list rather than a fixed significance threshold, and is considerably more robust to per-gene noise at small sample sizes than single-gene DE testing, which is why it was adopted for the low- n PCOS comparisons following the diagnosis in Section 4.6. Pathways were compared pairwise and three-way across datasets using a relaxed significance threshold ($q < 0.25$), standard practice in exploratory, hypothesis-generating GSEA

comparisons, with concordance defined as matching sign of normalized enrichment score (NES) between datasets.

5 Results

5.1 Brain-based classifier: valid model, no PCOS signal

After ComBat correction, the underlying AD vs. Control classifier achieved strong, properly cross-validated performance (CV AUC = 0.963, accuracy = 92.3%; Figure 1), ruling out classifier quality as an explanation for any downstream result. Despite this, projected PCOS samples showed no direction-correct “AD-likeness” signal: mean score for PCOS samples was 0.72, compared to 0.85 for Control samples (Figure 2) – the opposite direction from the original hypothesis, though with substantial overlap between groups given $n = 7$ (PCOS) vs. $n = 6$ (Control).

5.2 Tissue-matched blood classifier: replication of the null result

The blood-based AD classifier (GSE63061 vs. GSE54248, both same-platform, ComBat-harmonized) was also valid under cross-validation, if more modest, at CV AUC = 0.755 (Figure 1) – closely matching the range (0.657–0.874) reported by Lee and Lee [4] using the same underlying AD cohort, providing external corroboration that this classifier is reasonably constructed. Projected PCOS-blood samples again showed no PCOS-favoring signal: mean AD-likeness was 0.43 for PCOS versus 0.56 for Control (Figure 2). This result was independently corroborated by the model-free nearest-centroid method (PCOS = 0.45, Control = 0.52).

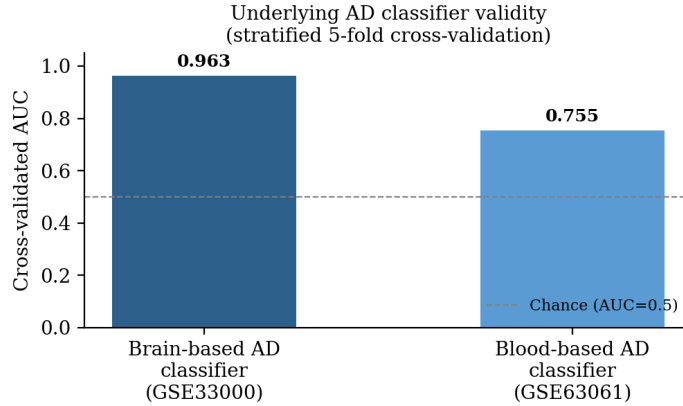


Figure 1: Cross-validated AUC for the underlying AD vs. Control classifiers in both tissue settings. Both substantially exceed chance, confirming classifier validity independent of the PCOS projection result.

Taken together, across two independent tissues, two harmonization runs, and two independent projection methods (a trained classifier and a training-free distance metric), no configuration produced a PCOS-favoring individual-level AD-likeness signal. This convergence substantially strengthens confidence that the null result reflects a genuine absence of individual-level signal in this data, rather than a methodological artifact specific to one tissue, platform, or method.

5.3 Pathway-level analysis: a replicated, divergent signal

In contrast to the individual-level null result, pathway-level GSEA revealed a specific, replicated pattern. Of 17 pathways enriched (at $q < 0.25$) in both independent PCOS datasets, 14 were concordant in direction between PCOS-adipose and PCOS-blood – despite being different tissues,

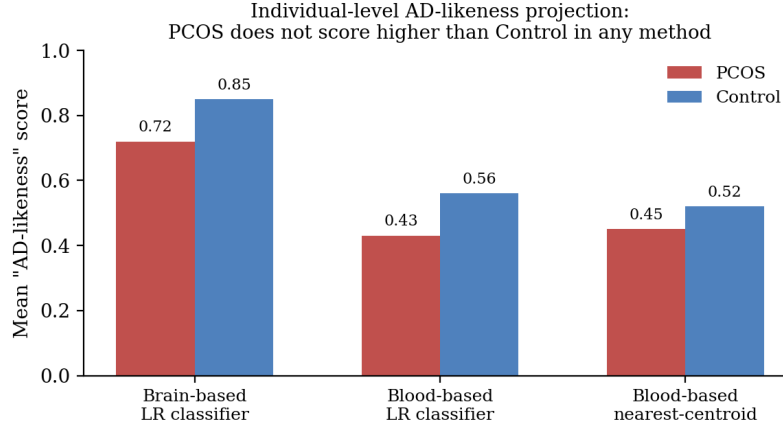


Figure 2: Mean individual-level “AD-likeness” score for PCOS vs. Control samples, across two tissues and two independent projection methods. In every case, PCOS scores equal to or lower than Control, opposite the direction the original hypothesis predicted.

different platforms (RNA-seq vs. microarray), and entirely independent subject cohorts. All 14 concordant pathways were immune- or inflammation-related, and all were *suppressed* in PCOS, while the same pathways were *elevated* in AD brain tissue (Table 2), the opposite pattern from a naive “shared inflammation” hypothesis.

Table 2: Representative concordant immune/inflammatory pathways: consistent direction across two independent PCOS cohorts (adipose, blood), opposite direction in AD brain.

Pathway	AD (brain)	PCOS (adipose)	PCOS (blood)
TNF- α /NF- κ B signaling	↑	↓	↓
IL6/JAK/STAT3 signaling	↑	↓	↓
Interferon- γ response	↑	↓	↓
Inflammatory response	↑	↓	↓
Complement	↑	↓	↓
Chemokine signaling	↑	↓	↓
Leukocyte transendothelial migration	↑	↓	↓

A small number of pathways (3 of 17) broke this pattern – notably allograft rejection, cell adhesion molecules, and hematopoietic cell lineage, which were concordant between AD and PCOS-blood but not PCOS-adipose – suggesting the divergence, while dominant, is not absolute across all immune-adjacent pathways. One additional finding, KEGG_RIBOSOME, showed strong, unusually large enrichment in both PCOS datasets independently (adipose NES= 3.94; blood NES= 2.25); because this pattern replicated across two different platforms and tissues, it is less likely to be a pure technical artifact of one experiment, but ribosomal/translation gene sets are a recognized source of confounding in expression studies generally, so this finding is reported with appropriate caution rather than as a confirmed biological result.

6 Discussion

The central methodological finding of this investigation is that a naive individual-level classification approach – train an AD classifier, project PCOS samples through it, interpret the score as risk – is fragile in at least two distinct ways that are easy to miss without deliberate stress-testing. First, cross-platform scale mismatches between training and projection data can produce a completely

uninformative, saturated output that superficially resembles a strong finding (all samples scoring near 1.0) rather than an obvious failure. Second, even after this is corrected, tissue mismatch between the classifier’s training data and the population being projected remains a distinct, separate confound that requires independent testing to rule out – correcting platform mismatch alone does not guarantee tissue mismatch is not still silently distorting results in one direction or another. Only by deliberately replicating the full pipeline in a tissue- and platform-matched setting, and further confirming with an independent, model-free method, could we distinguish a genuine absence of signal from an uncontrolled artifact.

The population-level finding – concordant immune and inflammatory pathway suppression in PCOS, contrasting with elevation in AD brain – is a more nuanced result than either a simple “PCOS causes AD via shared inflammation” hypothesis or a null finding would suggest. One plausible interpretation is that PCOS involves a distinct form of immune dysregulation (blunted or altered innate immune signaling) rather than simple hyperinflammation, and that this dysregulation could, speculatively, impair the body’s later capacity to mount an appropriate neuroinflammatory response to emerging AD pathology, rather than directly driving neurodegeneration through a shared inflammatory pathway. This is consistent with, though distinct from, findings in Lee and Lee [4], who found AD-associated blood gene signatures enriched for immune and inflammatory pathways broadly consistent with brain-tissue AD studies – our finding adds that the same broad pathway categories move in the *opposite* direction in PCOS, a comparison not previously reported to our knowledge.

7 Limitations

Several limitations should be considered when interpreting these results.

1. **Small PCOS sample sizes.** Both PCOS cohorts (adipose $n = 13$, blood $n = 8$) are small even by the standards of PCOS transcriptomics, a field where individual public datasets are typically $n < 20$ [3]. While cross-dataset replication increases confidence, it does not eliminate the statistical power limitations of either dataset individually.
2. **Cross-sectional, not longitudinal, design.** All datasets used here are cross-sectional. Even a hypothetically perfect classifier trained on this data would measure molecular *similarity* between PCOS and AD at a single point in time, not future *risk* of AD onset. Establishing true risk prediction would require longitudinal cohorts tracking PCOS patients over years or decades for AD outcomes, which does not currently exist in public repositories to our knowledge.
3. **ComBat harmonization assumption.** ComBat harmonization between the AD and PCOS datasets was performed without a shared biological covariate to preserve, since AD/Control status and PCOS/Control status are not common labels across the two studies. This assumes non-batch biological variation is reasonably exchangeable across the two studies, an assumption that cannot be fully verified and is a known soft spot of this kind of cross-study harmonization.
4. **Substantial gene loss during harmonization.** Restricting to genes with complete data across both platforms in the brain-adipose comparison retained only 1,261 of 9,180 originally common genes (13.7%), due to missing values in the processed GSE33000 matrix. It is possible, though not established, that the excluded genes were non-randomly distributed with respect to relevant biological pathways.
5. **Ribosome pathway signal requires independent confirmation.** As noted in Section 5, the strong ribosomal pathway enrichment in PCOS, while replicated across platforms, is a recognized category of technical confound in expression studies and should not be treated as a confirmed biological finding without further validation.

6. **No independent AD-side replication.** While the PCOS side of the analysis was replicated across two independent cohorts, the AD side used only one dataset per tissue (GSE33000 for brain, GSE63061 for blood). A fully symmetric replication design would also test AD-side reproducibility using a second independent AD cohort per tissue.

8 Future Work

Several concrete extensions follow directly from the limitations above. First, following standard practice in the PCOS transcriptomics literature [3], future work could pool additional small PCOS cohorts (e.g., ovarian/granulosa cell datasets such as GSE138518) with appropriate cross-study batch correction, increasing statistical power beyond what any single available dataset provides, while extending the tissue-replication design to a third tissue type. Second, leading-edge gene analysis on the concordant pathways identified in Section 5 would allow the discussion to move from pathway-level categories to specific candidate genes, strengthening the mechanistic specificity of the finding. Third, and most importantly for the original clinical motivation of this work, genuine progress on individual-level AD risk prediction in PCOS patients will require longitudinal cohort data linking PCOS diagnosis to later AD outcomes, which is a substantially larger data-collection undertaking beyond the scope of reanalyzing existing cross-sectional public datasets.

9 Conclusion

This investigation set out to test whether PCOS gene expression could signal individual-level Alzheimer’s disease risk from public transcriptomic data. Through a deliberately rigorous process of failure diagnosis, correction, and independent multi-tissue, multi-method replication, we found consistent evidence that it currently cannot: a validated, properly cross-validated classifier detected no PCOS-favoring signal in either brain-adipose or blood-blood tissue-matched comparisons, using either a trained model or a model-free distance metric. This negative result, achieved only after ruling out two distinct and easy-to-miss confounds (platform mismatch and tissue mismatch), is itself a meaningful methodological contribution given how easily a less rigorous version of this analysis could have produced a misleading positive or negative result. Separately, population-level pathway analysis revealed a specific, replicated, and biologically interpretable pattern – concordant immune and inflammatory pathway suppression in PCOS, contrasting with elevation in AD brain – that represents a more tractable and better-supported direction for future mechanistic and longitudinal research than individual-level risk classification given current public data constraints.

Code and Data Availability

All analysis code, intermediate outputs, and figures for this investigation are maintained in a private GitHub repository, available upon request: <https://github.com/rozakter18> (repository name withheld here; contact the author for access). All datasets used (GSE33000, GSE84958, GSE54248, GSE63061) are publicly available from the NCBI Gene Expression Omnibus.

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